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INT 7031 - Web Mining

Lecture 1: Introduction to Web Mining

Hoang-Quynh Le (PhD)

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Outline

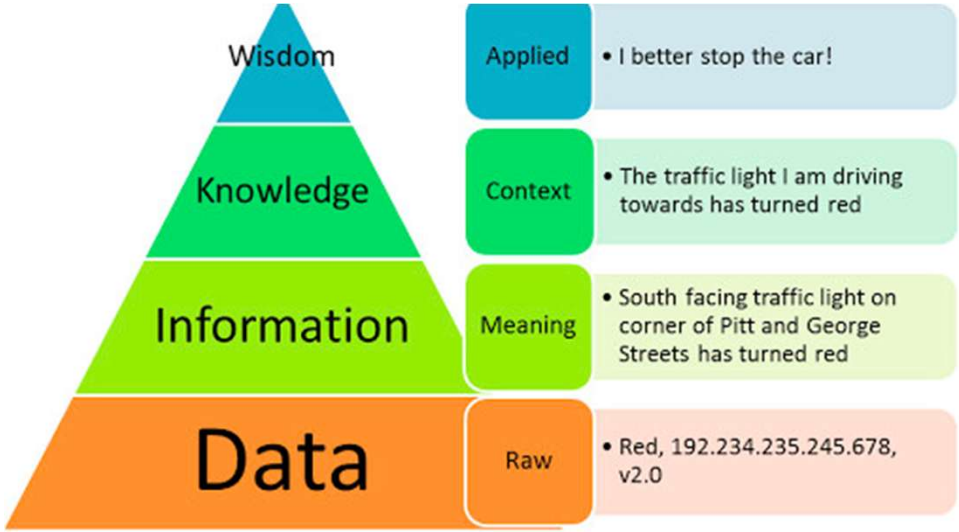
- Data Mining
  - Why Data Mining?
  - What is Data Mining?
  - Knowledge Discovery Process
  - Applications of Data Mining
- Web Mining
  - What is Web Mining
  - Types of Web Mining

Data Never Sleeps

Source: https://www.domo.com/learn/infographic/data-never-sleeps-11

https://youtu.be/dgY5wGHEgXE

From Data to Wisdom



- **Data**
  - The raw material of information
- **Information**
  - Data organized and presented by someone
- **Knowledge**
  - Information read, heard or seen and understood and integrated
- **Wisdom**
  - Distilled knowledge and understanding which can lead to decisions

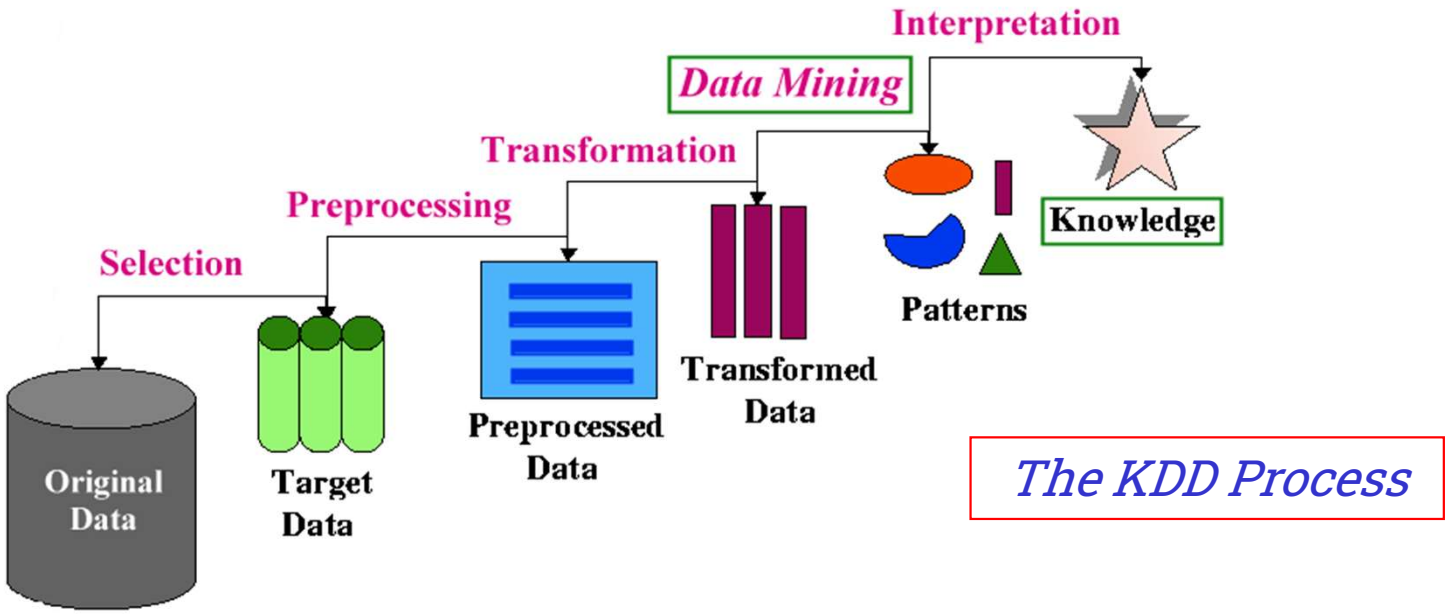
What is Data Mining?

The *non-trivial* extraction of *implicit*, *previously unknown* and potentially *useful* knowledge from data in large data repositories

- **Non-trivial**: obvious knowledge is not useful
- **implicit**: hidden difficult to observe knowledge
- **previously unknown**
- **potentially useful**: actionable; easy to understand

The Knowledge Discovery Process

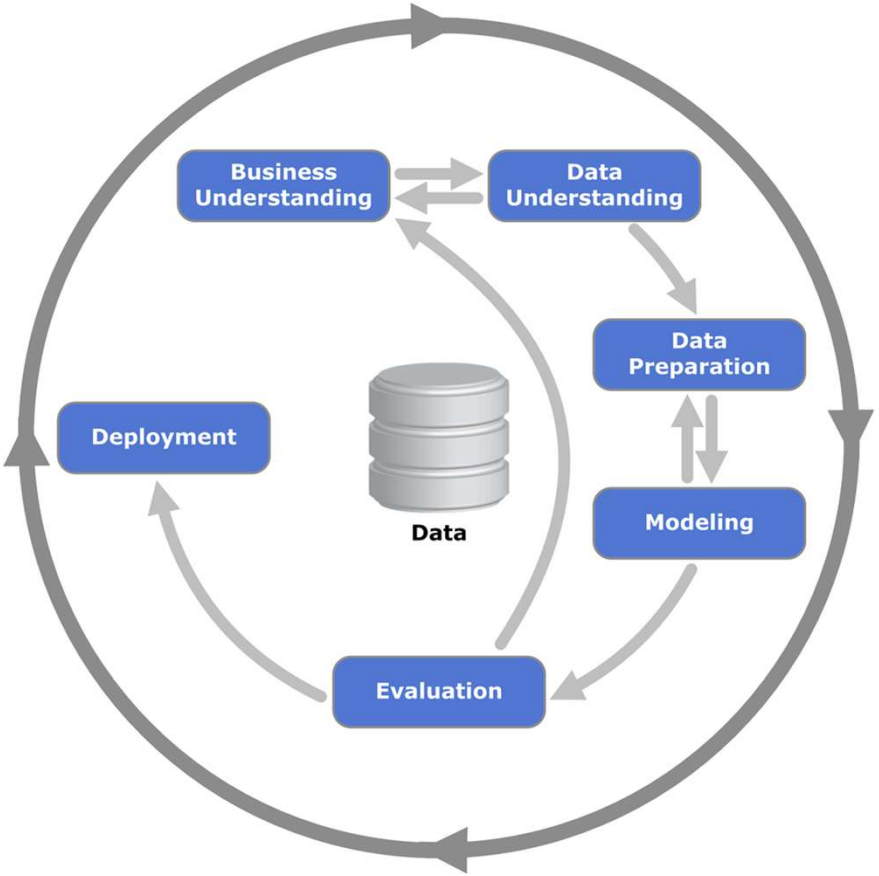
- **Data Mining v. Knowledge Discovery in Data (KDD)**
  - DM and KDD are often used interchangeably
  - actually, DM is only part of the KDD process



Types of Knowledge Discovery

- **Directed Knowledge Discovery**
  - Purpose: Explain value of some field in terms of all the others (goal-oriented)
  - Method: select the target field based on some hypothesis about the data; ask the algorithm to tell us how to predict or classify new instances
  - Examples:
    - what products show increased sale when cream cheese is discounted
    - which banner ad to use on a web page for a given user coming to the site
- **Undirected Knowledge Discovery**
  - Purpose: Find patterns in the data that may be interesting (no target field)
  - Method: clustering, affinity grouping
  - Examples:
    - which products in the catalog often sell together
    - market segmentation (find groups of customers/users with similar characteristics or behavioral patterns)

From Data Mining to Data Science



Data Mining: On What Kinds of Data?

- **Database-oriented data sets and applications**
  - Relational database, data warehouse, transactional database
  - Object-relational databases, Heterogeneous databases and legacy databases
- **Advanced data sets and advanced applications**
  - Data streams and sensor data
  - Time-series data, temporal data, sequence data (incl. bio-sequences)
  - Structure data, graphs, social networks and information networks
  - Spatial data and spatiotemporal data
  - Multimedia database
  - Text data and other semi-structured data
  - The World-Wide Web

Data Mining: On What Kinds of Data?

- **Structured Databases**
  - relational, object-relational, etc.
  - can use SQL to perform parts of the process

e.g., SELECT count(*) FROM Items WHERE type=video GROUP BY category

Borrow

customerID	date	itemID	#	…
C1234	99/09/06	98765	1	…
. . .				

Customer

customerID	name	address	password	birthdate	family_income	group	…
C1234	John Smith	120 main street	Marty	1965/10/10	$45000	A	…
. . .							

Items

itemID	type	title	media	category	Value	#	…
98765	Video	Titanic	DVD	Drama	$15.00	2	…
. . .							

Data Mining: On What Kinds of Data?

- **Flat Files**
  - most common data source
  - can be text (or HTML) or binary
  - may contain transactions, statistical data, measurements, etc.
- **Transactional databases**
  - set of records each with a transaction id, time stamp, and a set of items
  - may have an associated "description" file for the items
  - typical source of data used in market basket analysis

Rentals

transactionID	date	time	customerID	itemList
T12345	99/09/06	19:38	C1234	{I2, I6, I10, I45 …}
. . .				

Data Mining: On What Kinds of Data?

- **Other Types of Databases**
  - legacy databases
  - multimedia databases (usually very high-dimensional)
  - spatial databases (containing geographical information, such as maps, or satellite imaging data, etc.)
  - Time Series Temporal Data (time dependent information such as stock market data; usually very dynamic)
- **World Wide Web**
  - basically a large, heterogeneous, distributed database
  - need for new or additional tools and techniques
    - agents to assist in browsing and filtering
    - Web content, usage, and structure (linkage) mining tools
  - The “social Web”
    - User generated metadata, social networks, shared resources, etc.

What can Data Mining do?

- **Many Data Mining Tasks**
  - often inter-related
  - often need to try different techniques/algorithms for each task
  - each tasks may require different types of knowledge discovery
- **What are some of data mining tasks**
  - Classification
  - Prediction
  - Clustering
  - Association discovery
  - Sequence Analysis
  - Characterization
  - Discrimination

Applications of Data Mining

- **Business data analysis and decision support**
  - Marketing focalization
    - Recognizing specific market segments that respond to particular characteristics
    - Return on mailing campaign (target marketing)
  - Customer Profiling
    - Segmentation of customer
    - Customer behavior understanding
    - Customer retention and loyalty
    - Mass customization / personalization

Applications of Data Mining

- **Business data analysis and decision support**
  - Market analysis and management
    - Provide summary information for decision-making
    - Market basket analysis, cross selling, market segmentation.
    - Resource planning
  - Risk analysis and management
    - "What if" analysis
    - Forecasting
    - Pricing analysis, competitive analysis
    - Time-series analysis (Ex. stock market)

Applications of Data Mining

- **Fraud detection**
  - Detecting telephone fraud:
    - Telephone call model: destination of the call, duration, time of day/week
    - Analyze patterns that deviate from an expected norm
  - Detection of credit-card fraud
  - Detecting suspicious money transactions (money laundering)
- **Text mining:**
  - Message filtering (e-mail, newsgroups, etc.)
  - Newspaper articles analysis
  - Text and document categorization
- **Web Mining**
  - Mining patterns from the content, usage, and structure of Web resources

What is Web Mining?

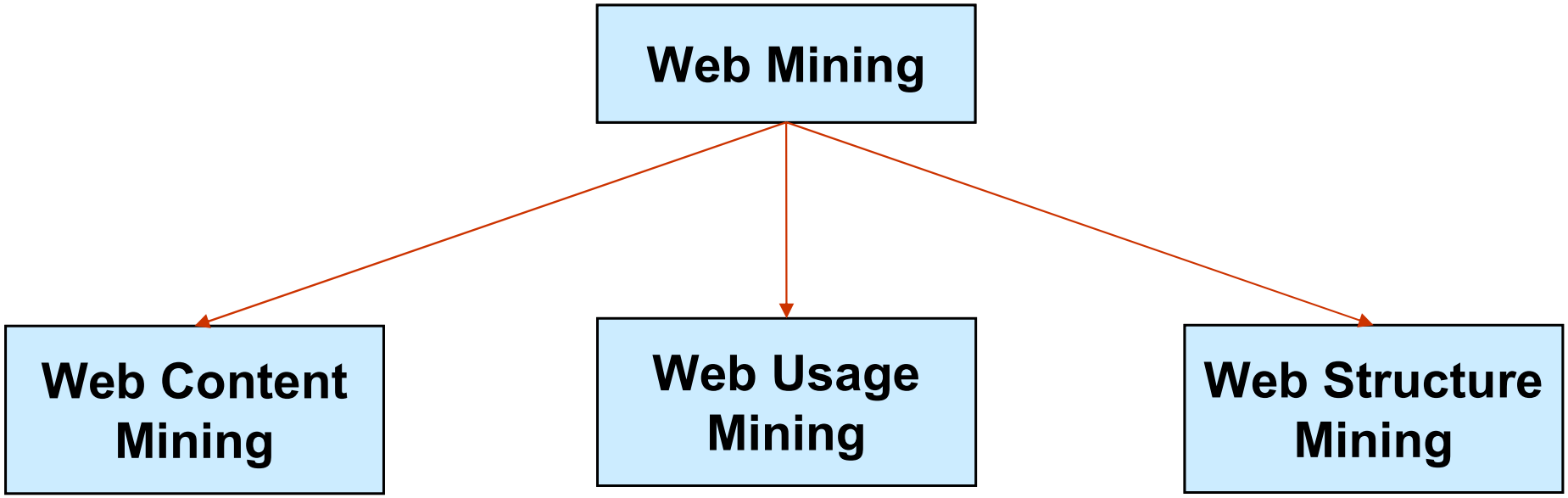
application of ***data mining*** and ***machine learning*** techniques to extract useful knowledge from **the content, structure, and usage of Web resources.**

But, why is this important and why is it more relevant than at any other time during the history of the Web?

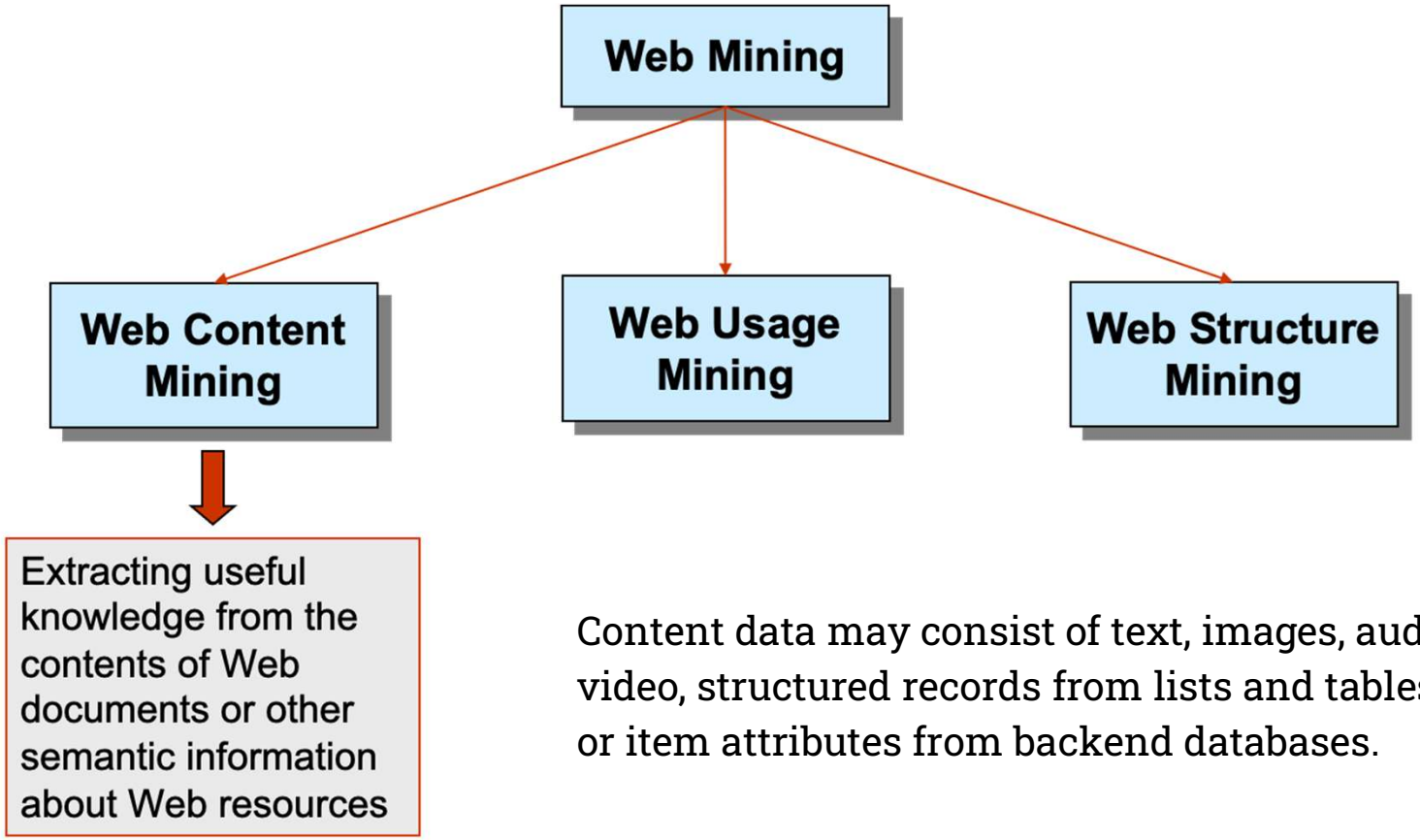
The New World of “Big Data” Internet

- **Leveraging big data**
  - Many of these applications manage, clean, and preprocess integrate often unstructured data from across many channels
  - Biggest challenge is in data distillation and preprocessing
- **Effective use of data mining and analytics**
  - No longer just a luxury but an integral part of systems
  - Especially important to leverage and effectively use user behavior and social data
- **Real-time deployment of models**
  - Needed for effective delivery of relevant, targeted, personalized content
  - Especially important on the Web: Predictive User Modeling

Types of Web Mining

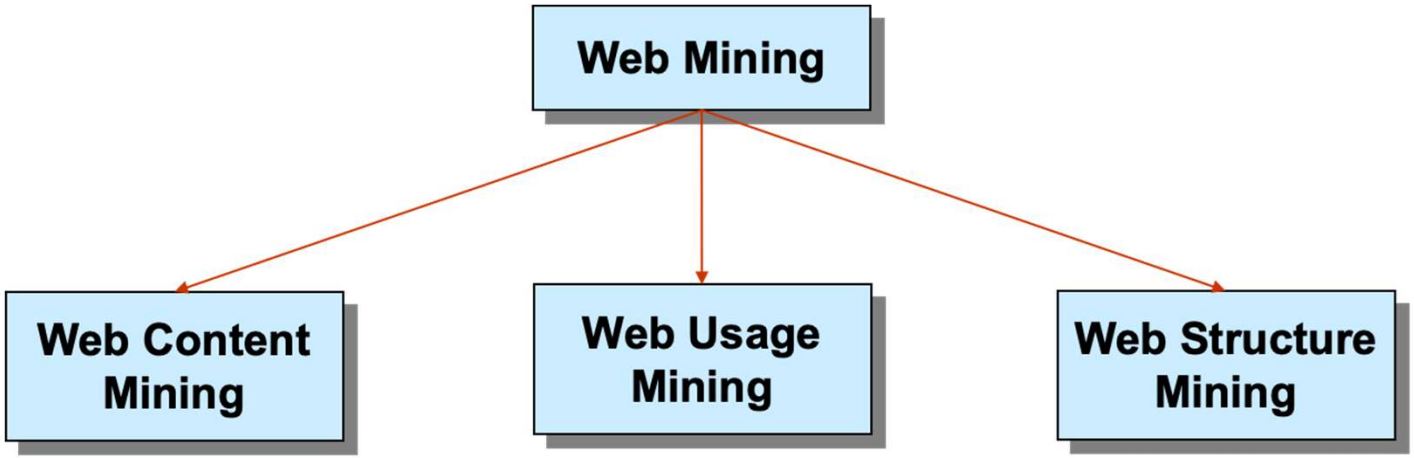


Web Content Mining



Content data may consist of text, images, audio, video, structured records from lists and tables, or item attributes from backend databases.

Web Content Mining

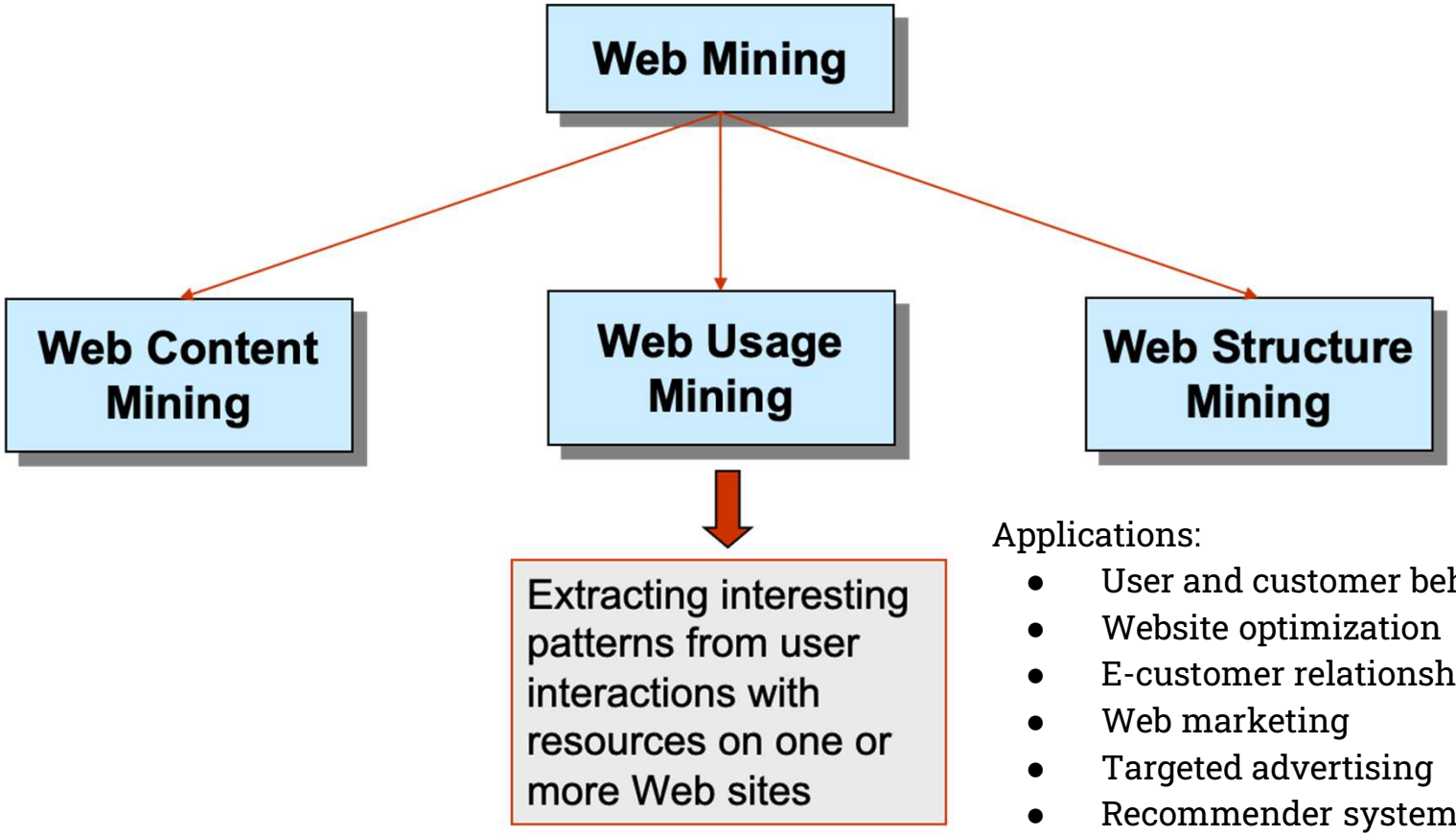


Extracting useful knowledge from the contents of Web documents or other semantic information about Web resources

Applications:

- document clustering or categorization
- topic identification / tracking
- concept discovery
- focused crawling
- content-based personalization
- intelligent search tools

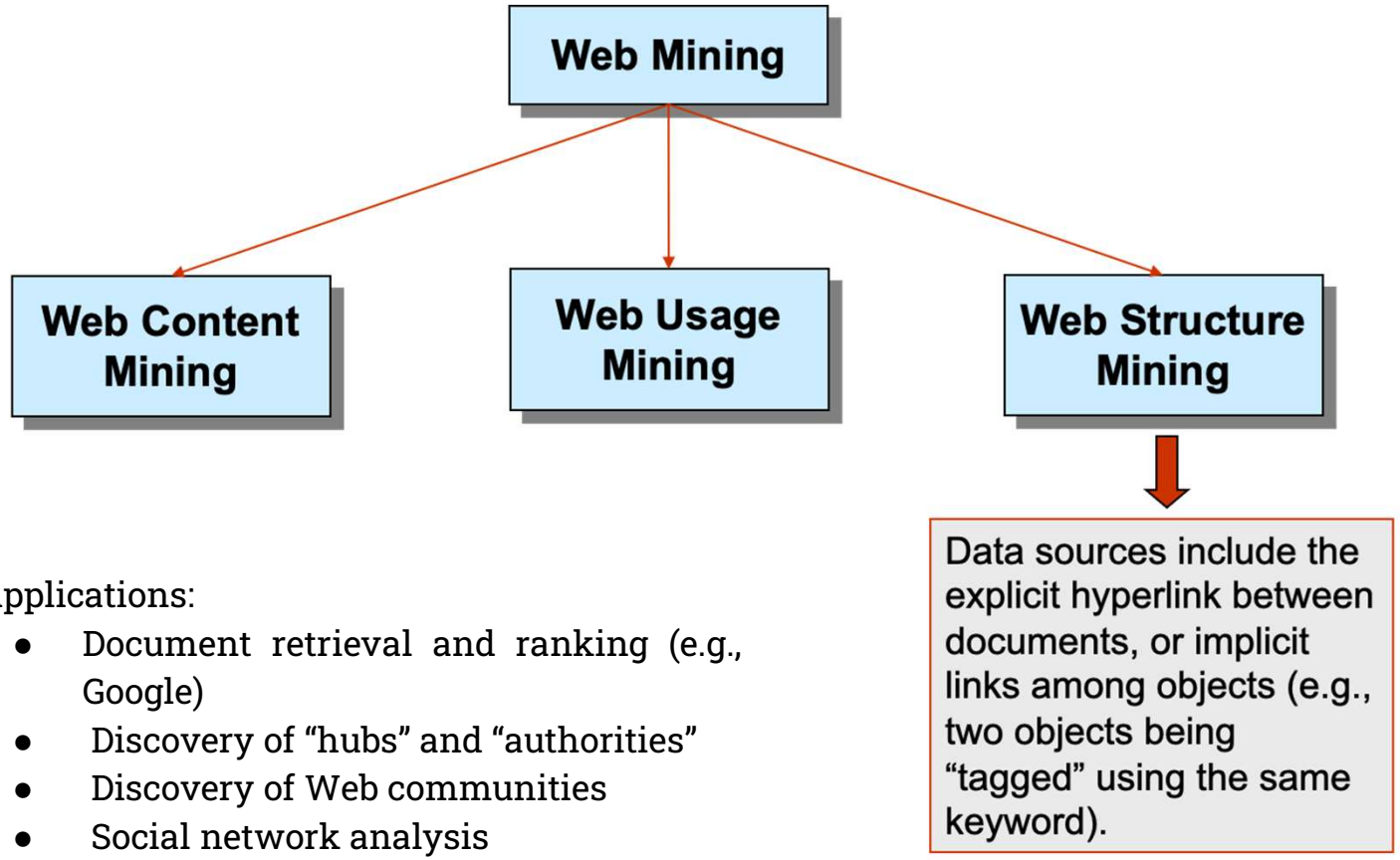
Web Usage Mining



Applications:

- User and customer behavior modeling
- Website optimization
- E-customer relationship management
- Web marketing
- Targeted advertising
- Recommender systems

Web Structure Mining



Applications:

- Document retrieval and ranking (e.g., Google)
- Discovery of "hubs" and "authorities"
- Discovery of Web communities
- Social network analysis

Summary

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  - Why Data Mining?
  - What is Data Mining?
  - Knowledge Discovery Process
  - Applications of Data Mining
- Web Mining
  - What is Web Mining
  - Types of Web Mining

Thank you

Email me
lhquynh@vnu.edu.vn